BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm
Duration: 1 hour 20 min

Semester: Fall 2023
Full Marks: 30

CSE 320: Data Communications

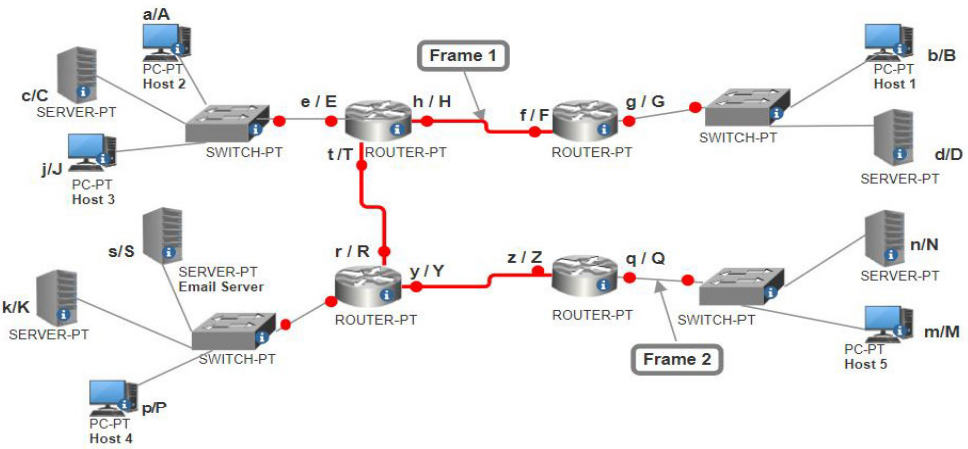
Answer the following questions.
Figures in the right margin indicate marks.

SET A

Name:	ID:	Section:

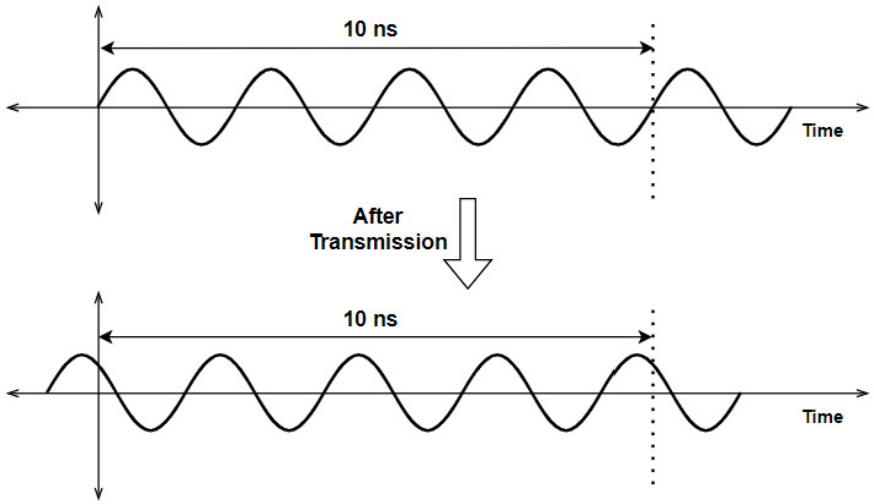
1. [CO1]

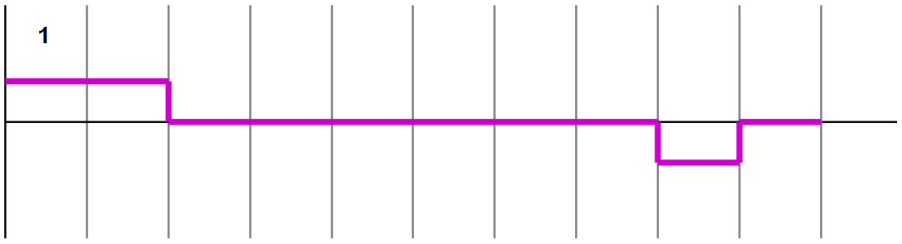
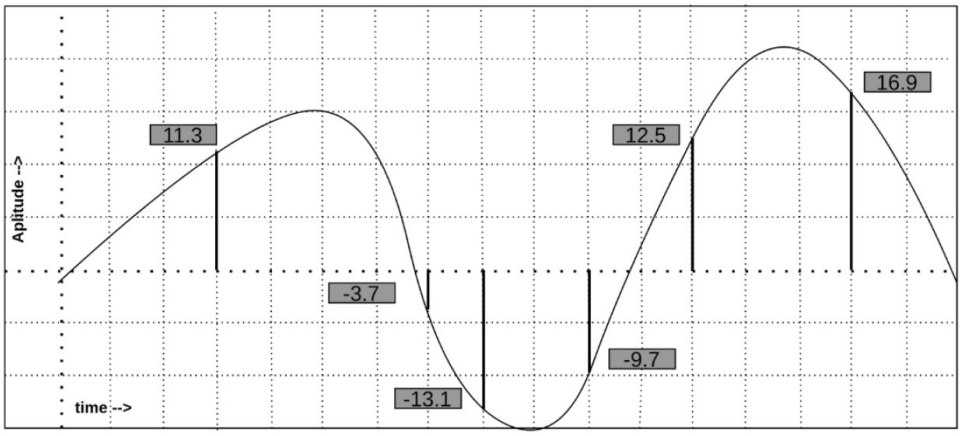
a) Complete the frames (1 & 2) given below with appropriate port, IP and MAC addresses. Frame 1 is intended for Host 5 for sending a message from Host 1. Frame 2 is coming from the Email server to Host 5. Assume that Mac addresses are Uppercase Letters, and logical addresses are lowercase letters. For Port Address, use Dynamic Port address, for the user processes (49,152 - 65,535), and for server, use well- known addresses (0-1023). [4]



Frame 1	Destination MAC	Source MAC	Destination IP	Source IP	Destination Port	Source Port	Data	Trailer
Frame 2	Destination MAC	Source MAC	Destination IP	Source IP	Destination Port	Source Port	Data	Trailer

b) Let there are 3 buildings in your university campus. Buildings are connected using mesh topology. Each building has 4 computer Labs. Each Lab has 8 computers. The computers are connected using bus topology but the Labs of a single building are connected using star topology. Draw your campus hybrid topology. How many links will there be? [4]

	c)	Suppose you want to send a letter to your friend. Find out which layer of the **OSI model** corresponds to the following tasks you are going to perform: • For secrecy, you write the letter using Caesar cipher. • Your building has multiple flats, but they use a common mailbox. You drop your letter to the mailbox. • Your letter is sorted and redirected to the post office of the receiver's area by your area's post office. • Your letter is carried by a postman from the mailbox of your building to the nearest post office.	[2]
2. [CO2]	a)	I. Find the frequency and time period of the given signal before transmission. II. Can you relate any kind of transmission impairment with the scenario in the given figure after transmission? Justify your answer. 	[2]
	b)	Assume, we have a channel with a 5 GHz bandwidth with 800 mW noise. If the signal strength is 36 W, what are the appropriate bit rate and signal level?	[3]
	c)	Suppose you want to send a pdf file containing 12 pages to your friend through a 5 Mbps optical fiber line. A page is an average of 25 lines with 80 characters. Assume that the distance between you and your friend is 8000 km. There are two routers in the transmission path and each has 5 ms of processing time on average. **Determine** the latency of the transmission. (Light travels at 2.8×10^8 m/s)	[5]

3. **[CO2]**	a)	The following signal is drawn using a multi-transition line encoding technique where 3 transition rules and 3 levels are used. i. Decode the bit stream from it. ii. Encode it using Bipolar AMI encoding, **apply** any appropriate Scrambling method to get rid of the consecutive 0 problem if there is any. 	[2+ 3]
	b)	The following figure depicts a sampled analog signal for digital signal representation. By applying the concept of **Pulse Code Modulation**, assume there will be **3-bit** code words for each sampled amplitude. **Show** the normalized **Quantized value** and **quantization code** and **Binary Encoded value** for the given analog signal value at different time stamps. Assume that the sampling amplitudes are between –15V to +25V. 	[5]